

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                              |
|-----------------------------|----------------------------------------------|
| <b>Product Name</b>         | <u>Oxalic acid, anhydrous</u>                |
| <b>CAS No</b>               | 144-62-7                                     |
| <b>Synonyms</b>             | Ethanedionic acid                            |
| <b>Molecular Formula</b>    | C <sub>2</sub> H <sub>2</sub> O <sub>4</sub> |
| <b>Molecular Weight</b>     | 90.04                                        |
| <b>Recommended Use</b>      | Laboratory chemicals.                        |
| <b>Uses advised against</b> | No Information available                     |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>O/0580/53</b>                                                                            |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number      HSR002491**

### GHS Classification

#### Physical hazards

Substances/mixtures corrosive to metal Category 1

#### Health hazards

|                                                      |              |
|------------------------------------------------------|--------------|
| Acute Oral Toxicity                                  | Category 4   |
| Acute Dermal Toxicity                                | Category 4   |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 4   |
| Skin Corrosion/Irritation                            | Category 1 C |
| Serious Eye Damage/Eye Irritation                    | Category 1   |
| /; Effects on or via lactation                       |              |
| Specific target organ toxicity - (repeated exposure) | Category 2   |

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



Signal Word

Danger

### Hazard Statements

H373 - May cause damage to organs through prolonged or repeated exposure

H290 - May be corrosive to metals

H314 - Causes severe skin burns and eye damage

H362 - May cause harm to breast-fed children

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

### Precautionary Statements

#### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P363 - Wash contaminated clothing before reuse

#### Storage

P403 - Store in a well-ventilated place

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other hazards which do not result in classification

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component   | CAS No   | Weight % |
|-------------|----------|----------|
| Oxalic acid | 144-62-7 | >95      |

## Section 4 - First Aid Measures

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### New Zealand Emergency Tel.

CHEMTREC®  
09 980 6780 or +64 9 980 6780

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Causes severe eye damage.                                                                                           |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Corrosive material. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

#### **Environmental Precautions**

Should not be released into the environment.

#### **Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture.

**Incompatible Materials**

Strong oxidizing agents. Strong bases. Metals. Acid chlorides.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## **Section 8 - Exposure Controls and Personal Protection**

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component   | New Zealand WEL                                       | Australia                                             | ACGIH TLV                                             | The United Kingdom                                                |
|-------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------|
| Oxalic acid | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material                                               | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|--------------------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Butyl rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Particulates filter conforming to EN 143 (or AUS/NZ equivalent)                                                                                                                                                                                                                                                             |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                 |                                          |
|------------------------------------------------|---------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Powder Solid                    |                                          |
| <b>Appearance</b>                              | White                           |                                          |
| <b>Odor</b>                                    | Odorless                        |                                          |
| <b>Odor Threshold</b>                          | No data available               |                                          |
| <b>pH</b>                                      | 1.3                             | 9 g/L                                    |
| <b>Melting Point/Range</b>                     | 189 - 191 °C / 372.2 - 375.8 °F |                                          |
| <b>Softening Point</b>                         | No data available               |                                          |
| <b>Boiling Point/Range</b>                     | No information available        |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available        |                                          |
| <b>Explosion Limits</b>                        | No data available               |                                          |
| <b>Flash Point</b>                             | > 93.4 °C / > 200.1 °F          | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | Not applicable                  |                                          |
| <b>Decomposition Temperature</b>               | No data available               |                                          |
| <b>Viscosity</b>                               | Not applicable                  | Solid                                    |
| <b>Water Solubility</b>                        | 90 g/L (20°C)                   |                                          |
| <b>Solubility in other solvents</b>            | No information available        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                  |                                          |
| Oxalic acid                                    | -1.7                            |                                          |
| <b>Vapor Pressure</b>                          | < 0.01 mmHg @ 20 °C             |                                          |
| <b>Density / Specific Gravity</b>              | 1.900                           |                                          |
| <b>Bulk Density</b>                            | No data available               |                                          |
| <b>Vapor Density</b>                           | Not applicable                  | Solid                                    |
| <b>Particle characteristics</b>                | No data available               |                                          |

### Other information

|                          |                        |
|--------------------------|------------------------|
| <b>Molecular Formula</b> | C2 H2 O4               |
| <b>Molecular Weight</b>  | 90.04                  |
| <b>Evaporation Rate</b>  | Not applicable - Solid |

## Section 10 - Stability and Reactivity

|                                  |                                                                |
|----------------------------------|----------------------------------------------------------------|
| Reactivity                       | No                                                             |
| Stability                        | Stable under normal conditions.                                |
| Sensitivity to Mechanical Impact | No information available                                       |
| Sensitivity to Static Discharge  | No information available                                       |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                       |
| Hazardous Reactions              | None under normal processing.                                  |
| Conditions to Avoid              | Avoid dust formation, Incompatible products, Excess heat.      |
| Incompatible Materials           | Strong oxidizing agents, Strong bases, Metals, Acid chlorides. |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).       |

## Section 11 - Toxicological Information

### Acute Effects

#### Information on likely routes of exposure

#### Product Information

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| Inhalation | Avoid breathing dust or spray mist. May be harmful if inhaled.                                  |
| Eyes       | Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. |
| Skin       | Avoid contact with skin. Skin Corrosion/Irritation. Harmful in contact with skin.               |
| Ingestion  | May be harmful if swallowed.                                                                    |

#### Numerical measures of toxicity

##### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Category 4                                                       |
| Dermal     | Category 4                                                       |
| Inhalation | Based on available data, the classification criteria are not met |

| Component   | LD50 Oral         | LD50 Dermal     | LC50 Inhalation |
|-------------|-------------------|-----------------|-----------------|
| Oxalic acid | 375 mg/kg ( Rat ) | 20 g/kg ( Rat ) |                 |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

|             |                                                                  |
|-------------|------------------------------------------------------------------|
| Respiratory | Based on available data, the classification criteria are not met |
| Skin        | Based on available data, the classification criteria are not met |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Category 2

**Target Organs** Liver, Kidney, Respiratory system, Eyes, Skin.

(j) aspiration hazard; Not applicable  
Solid

**Symptoms / effects, both acute and delayed**  
No information available.

## Section 12 - Ecological Information

### Ecotoxicity

#### Aquatic ecotoxicity

| Component   | Freshwater Fish | Water Flea            | Freshwater Algae | Microtox |
|-------------|-----------------|-----------------------|------------------|----------|
| Oxalic acid |                 | EC50 = 136.9 mg/L/48h |                  |          |

**Terrestrial ecotoxicity** There is no data for this product

#### Persistence and Degradability

**Persistence** Persistence is unlikely.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Oxalic acid | -1.7    | No data available             |

**Mobility** The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### Other adverse effects

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Solutions with low pH-value must be neutralized before discharge.

## Section 14 - Transport Information

**NZS 5433:2020** Not regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR002491 |
|-----------------------------|-----------|

**National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions** Not applicable



according to EU REACH

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component   | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| Oxalic acid | 144-62-7 | X     | X    | 205-634-3 | -      | -   | KE-13152 | X     | X    |

| Component   | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-------------|----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Oxalic acid | 144-62-7 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**Section 16 - Other Information**

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

**Legend**

NZIoC - New Zealand Inventory of Chemicals

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

NZS 5433:2020 - Transport of Dangerous Goods on Land

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

AICS - Australian Inventory of Chemical Substances

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists

PNEC - Predicted No Effect Concentration

OECD - Organisation for Economic Co-operation and Development

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date**

13-Mar-2023

**Revision Summary**

Not applicable

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**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**